Remote Sensing of Environment 325 (2025) 114775

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

A vision foundation model-based method for large-scale forest disturbance mapping using time series Sentinel-1 SAR data

Yuping Tian [a], Feng Zhao [a,*], Ran Meng [b,c], Rui Sun [d], Yuan Zhang [a], Yanyan Shen [d], Bin Wang [a], Jie Liu [b,c], Mingze Li [a,*]

[a] Key Laboratory of Sustainable Forest Ecosystem Management-Ministry of Education/College of Forestry, Northeast Forestry University, Harbin 150040, China
[b] Artificial Intelligence Research Institute, Faculty of Computing, Harbin Institute of Technology, Harbin 150006, China
[c] State Key Lab of Smart Farm Technologies and Systems, Harbin Institute of Technology, Harbin 150080, China
[d] College of Resources and Environment, Huazhong Agricultural University, Wuhan 430070, China

ARTICLE INFO

Editor name: Dr. Marie Weiss

Keywords:
Pan-tropical deforestation
Image segmentation
C-band SAR
Large-scale model transfer
Improved greedy sampling

ABSTRACT

Accurate and timely forest disturbance mapping at large-scale is crucial for ecosystem protection and management. Sentinel-1 SAR data, with its all-weather capability and fine spatial-temporal resolutions, offers unique advantages for timely mapping of forest disturbance. Although deep learning models have been used for this purpose, they still struggle to fully exploit Sentinel-1 data's potential due to challenges in extracting multi-scale features and capturing context in complex landscape patterns. The advent of vision foundation models like the Segment Anything Model (SAM) offer new possibilities to improve large-scale forest disturbance mapping with Sentinel-1 data. However, trained with natural images, SAM had difficulty processing speckle noises in SAR and recognizing intricate forest disturbance patterns; additionally, challenges remained in developing model transfer strategies and improving model efficiency for large-scale applications. To address these issues, we propose SAMSR, a SAM-based framework adapted for forest disturbance mapping using time series Sentinel-1 data by adding CNN branches and cross-branch attention modules to enhance fine spatial details. Firstly, we evaluated SAMSR's performances against SAM, U-Net and DeepLabv3+ at four global disturbance hotspots with large variations in environmental conditions (Rondônia in Brazil, Guangxi in China, California in USA, and Hainan in China). Then, we test the model's transferability using fine-tuned transfer learning to identify the best transfer strategy and explore the potential of active learning to further enhance model efficiency. The results indicated that the SAMSR (IoU 0.50–0.78; F1 0.67–0.88) outperformed SAM, U-Net and DeepLabv3+ for about 23.08 %, 3.23 % and 1.59 % in IoU, respectively, while the multi-region model achieved optimal transfer performance (IoU 0.60–0.77; F1 0.75–0.87). Moreover, applying active learning can meet the saturation accuracy of traditional method with about 20 %–70 % less training samples, significantly reducing costs for fine-tuning the foundational model. This study thus provides a novel and efficient framework for large-scale forest disturbance monitoring at fine spatial-temporal resolutions, which could be critical for forest protection and ecological studies.

1. Introduction

Large-scale forest disturbance causes severe biodiversity loss and significant carbon dioxide emissions, thereby intensifying the greenhouse effect and destabilizing the global climate (Altarez and Maraseni, 2023; Brockerhoff et al., 2017; Miura et al., 2015). Timely and accurate monitoring of disturbance is thus crucial, as it can assist relevant agencies in promptly detecting and addressing illegal logging activities, and help assess and adjust forest management strategies through better understanding its ecological consequences on biodiversity, hydrological and carbon cycles (Decuyper et al., 2022; Griffiths et al., 2018; Wang et al., 2019). However, traditional field survey method is time- and labor-consuming, and thus cannot meet the need of forest disturbance mapping at fine spatial-temporal resolution.

Recent advances in remote sensing technology have enabled large-scale, accurate forest disturbance mapping (De Bem et al., 2020;

* Corresponding authors.
E-mail address: fzhao@nefu.edu.cn (F. Zhao).

https://doi.org/10.1016/j.rse.2025.114775
Received 24 September 2024; Received in revised form 5 April 2025; Accepted 20 April 2025
Available online 29 April 2025
0034-4257/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Hethcoat et al., 2021; Jamali et al., 2023; Solorzano et al., 2023). Optical data served as one of the main sources for forest disturbance mapping, detecting disturbance activity through pixel-based or object-based image classification, as well as multi-temporal change detection algorithms based on time series images (Chen et al., 2021; Grecchi et al., 2017; Hethcoat et al., 2019). While optical data is vital for global forest monitoring, its effectiveness is hindered by clouds and shadows, limiting its capability for timely forest disturbance mapping, especially in cloud-prone tropical areas (Alshehri et al., 2024; Cortez et al., 2024; Hansen et al., 2013). Techniques such as multi-temporal synthesis, gap-filling, and multi-sensor data fusion can potentially overcome cloud cover limitations by using cloud-free observations from different times or by combining data from multiple sensors (Qiu et al., 2023; Wang et al., 2021a, 2021b; Zhou et al., 2021). While these technologies represent a promising direction, they also introduce some challenges. For example, multi-sensor fusion increases computational costs and data processing complexity, and may introduce uncertainties due to temporal misalignment or inconsistent spatial resolutions (Ghamisi et al., 2019).

In contrast, Synthetic Aperture Radar (SAR) can penetrate clouds, offering more consistent opportunities for frequent disturbance mapping. Sentinel-1, with its all-weather capability and short revisit interval, is particularly suited for continuous forest monitoring in cloud-prone regions (Ballère et al., 2021; Hethcoat et al., 2021). Many studies have utilized SAR data for forest disturbance mapping, primarily relying on traditional approaches such as threshold-based methods and probabilistic detection method, which involve setting predefined thresholds for disturbance detection (Li et al., 2022a, 2022b; Reiche et al., 2018). Recently, more and more studies have highlighted the advantages of deep learning models, like U-Net, for improving the timeliness and accuracy of forest disturbance mapping using Sentinel-1 SAR data, as they can automatically learn complex features and capture disturbance patches across various shapes and orientations, outperforming traditional methods (Kattenborn et al., 2021; Solorzano et al., 2023; Zhao et al., 2022). However, they still face challenges when applied to Sentinel-1 data for forest disturbance mapping. Sentinel-1's SAR imagery is prone to speckle noises, which complicate the extraction of clear and reliable features necessary for accurate analysis, posing difficulties when dealing with the specific multi-scale complexities of forest environments. Additionally, in terms of contextual awareness, the model's reliance on local receptive fields can limit its ability to capture broader contextual information within the vast, heterogeneous forest landscapes across various areas.

With advances in computer vision, foundational image segmentation models tailored for remote sensing data have emerged, open up new avenues for improving large-scale forest disturbance mapping. Recently, Meta AI introduced the Segment Anything Model (SAM), a powerful computer vision foundation model trained on 11 million annotated natural images, capable of accurate segmentation in complex scenarios (Kirillov et al., 2023). A series of studies have combined optical remote sensing data with the SAM for land cover segmentation (Zhou et al., 2024) and crop type classification (Song et al., 2024), demonstrating higher mapping accuracy and highlighting the advantage of SAM in these tasks. The emergence of SAM provides new insights for further improving forest disturbance mapping based on remote sensing data.

However, integrating a foundational model like SAM with Sentinel-1 SAR data for forest disturbance mapping is not straightforward. The SAM was trained on a large-scale RGB images dataset and demonstrated excellent performance on segmentation tasks. Optical imagery, such as RGB images, captures reflected sunlight to provide rich spatial and spectral information that is crucial for accurate object identification and segmentation (Liu et al., 2024). This abundance of spectral and textural details allows models like SAM to effectively distinguish and delineate various objects with high precision. However, given the obvious differences between optical and SAR images, the SAM does not perform well on SAR images (Yan et al., 2023). SAR images rely on active microwave signals, which showed prominent speckle noises and reduced texture features, with varying backscatter characteristics responding to surface roughness, dielectric constant, etc., and making it difficult for SAM, pretrained on optical imagery, to effectively identify and segment targets (Singh and Shree, 2016; Yan et al., 2023). Moreover, the SAM was designed for versatility and breadth of use, despite of its overall well performance, experiments showed that it may miss the fine structure and cannot generate the boundary clearly and accurately for zero-shot transfers (Kirillov et al., 2023). Therefore, it is necessary to fine-tune or revise the SAM to enable forest disturbance monitoring based on SAR data.

To achieve this goal, it is crucial to enhance the SAM's ability to capture finer features and improve segmentation accuracy for blurry boundaries, complex shapes, and small-sized objects. The SAM divides the input image into non-overlapping 16×16 tokens before feature modeling, compromising crucial local information needed to identify small targets and boundaries in S1 images, which complicates accurate segmentation of disturbance patches with low contrast, weak boundaries, and complex shapes. Fortunately, CNN can effectively extract details and edge information from images through local convolutional operations and automatically learn local features at multiple scales, which can significantly enhance the segmentation capability for small targets and complex shapes (Liu et al., 2022; Ngo et al., 2024). Moreover, the effective integration of local and global features is crucial for improving model performance. The cross-branch attention mechanism establishes information flow between global and local features, effectively incorporating local features to complement the deficiencies of global features, thereby enhancing the segmentation performance and generalization ability of SAM on SAR images (Lin et al., 2023). Recent studies show that incorporating adapters into the basic SAM structure and adding CNN branches can help suppress noises and enhance extraction of fuzzy boundaries for small objects in ultrasound images (Lin et al., 2023; Tu et al., 2024), yet the role of this approach in forest disturbance monitoring and mapping based on Sentinel-1 data still requires further evaluation.

Despite the promising advancements brought by SAM and its adaptations, several additional challenges still exist for large-scale disturbance mapping at fine spatial-temporal resolution. Firstly, deep learning models require extensive high-quality annotated data for training, which is costly and time-consuming to acquire (Ahmed et al., 2023; Wang et al., 2021a, 2021b). Secondly, the heterogenous topography, species composition, and deforestation patterns make it difficult for models to transfer across different regions. Ideally, the combination of transfer and active learning techniques with the adapted SAM can save computational resources while maximizing the use of limited labeled data, offering a promising solution for addressing the challenges of efficiency and cost in large-scale mapping (Pan and Yang, 2009; Wu et al., 2019). On one hand, transfer learning allows models to apply pretrained knowledge to new tasks, reducing the need for labeled data and enhancing adaptability across regions (Ma et al., 2024a, 2024b; Martins et al., 2022; Zhu et al., 2024). Fine-tuning with local data is one of the most used transfer learning techniques: it can reduce the need for large amounts of labeled data, and enhance the model's transferability across different geographic regions and forest types (Vega et al., 2021; Zhao et al., 2022). On the other hand, active learning addresses those issues by optimizing the training process. Specifically, it can select the most representative samples and avoid redundancies to minimize the cost while optimize model performance (Cao et al., 2020; Tuia et al., 2011; Wan et al., 2023; Wu et al., 2019).

Therefore, in this study, we propose a novel framework called SAMSR with time series Sentinel-1 data to map large-scale forest disturbance across four pan-tropical hotspot areas. Here, forest disturbance was mainly referred to stand-replacing disturbance events. Our specific objectives include: (1) Evaluate the spatial and temporal performances of our proposed SAMSR, with the SAM, U-Net and DeepLabv3+ models as baselines. (2) Based on transfer learning methods, determine the optimal transfer strategies of our developed framework

across four large test areas with variations in environmental conditions (Rondônia in Brazil, Guangxi in China, California in USA, and Hainan in China). (3) Evaluate the effectiveness of active learning methods in reducing the required number of samples for model training and enhancing the efficiency of SAMSR.

2. Study area and materials

2.1. Study area

The proposed framework was implemented across four study areas, characterized by varying topography and forest disturbance patterns. The first area, in Rondônia, Brazil, spans 10,100 km^2 within the tropical rainforest, a key area for large-scale Amazon deforestation since the 1960s, with elevations ranging from 90 to over 400 m (Fig. 1a). The

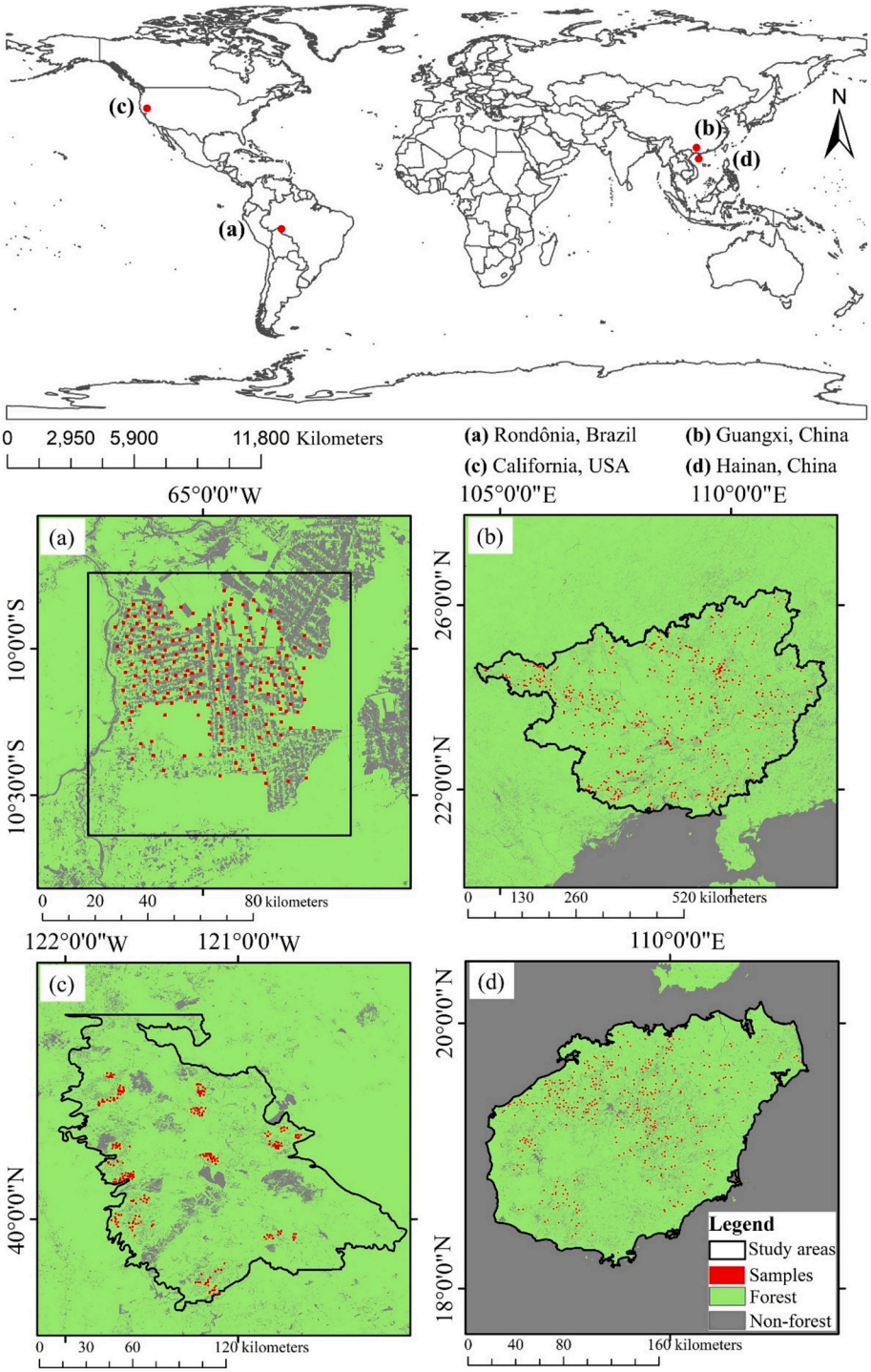


Fig. 1. Study areas in (a) Rondônia, Brazil, (b) Guangxi, China, (c) California, USA, and (d) Hainan, China. The red areas represent the sample of the study area utilized during model development, calibration, and evaluation. The green (forest) and grey (non-forest) areas are based on data from the 2015 Hansen Global Forest Change product. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

second area, in Guangxi, China, covers 236,700 km^2 with elevations from 10 to over 2000 m, where forests primarily face disturbance pressures on eucalyptus plantations driven by economic demands, which could lead to significant ecological issues (Fig. 1b). The third area, in northern California, USA, spans 16,500 km^2 and is a crucial timber management region with elevations ranging from 200 to over 3000 m (Fig. 1c). The fourth area, in Hainan, China, spans 35,400 km^2 with elevations from 0 to over 1800 m and contains the country's most concentrated and largest contiguous tropical rainforest, a vital part of the world's tropical ecosystems, crucial for biodiversity and ecological stability (Fig. 1d).

2.2. Reference data

High-resolution satellite images enabled the clear identification of disturbance patches, from which we collected reference data for subsequent use in model training, parameterization, and testing (Table 1). Specifically, we utilized 3-m resolution Planet images, collected between January 2016 and January 2023. In the four study areas, we generated stratified point locations with uniform spatial and temporal distribution, and ensured enough true positive and true negative samples. Each 1280 m × 1280 m square plot surrounding the selected point (128 × 128 pixels) served as the basic spatial unit for ground reference labeling and dataset segmentation.

2.3. Sentinel-1 data

Level-1 GRD Sentinel-1 data from the European Space Agency was utilized for disturbance mapping, with pre-processing steps include the application of orbit files, the removal of GRD border and thermal noise, followed by radiometric calibration, and terrain correction. Given the sensitivity of Sentinel-1 data to terrain variations, the standardized level-1 GRD data were further refined using 30-m SRTM DEM data to ensure accuracy in terrain relief representation (Zhao et al., 2022).

3. Methods

In this study, we proposed a novel approach (Fig. 2) for improved large-scale disturbance mapping using Sentinel-1 SAR data. The approach included three parts: data preprocessing, model development, and postprocessing. First, the pre-processed Sentinel-1 and Planet data were divided into training, validation and test data for subsequent model evaluation. Second, the SAMSR model was constructed based on the characteristics of the SAM and Sentinel-1 data. Third, transfer learning technology was used to test the transferability of the SAMSR, and evaluate the effectiveness of active learning to further improve the efficiency of the model. Finally, the spatiotemporal accuracy of the disturbance products was assessed through independent monthly validation using 3-m resolution Planet images.

Table 1
Total number of plots in the study area and number of plots used for training, parameterization and testing. PAR represents the perimeter area ratio, Q1 represents the lower quartile, M represents the median, and Q3 represents the upper quartile.

Study areas	Total sample plots	Training/ Parameterization/ Test plots	PAR (Q1/M/ Q3)	Sample elevation range (m)	Sample slope range (degree)
Rondônia (Ron)	198	151 / 24 / 23	0.0103/ 0.0122/ 0.0138	161.16 ± 20.20	1.02 ± 0.43
Guangxi (GX)	495	399 / 49 / 47	0.0235/ 0.0300/ 0.0326	378.41 ± 250.03	8.21 ± 4.20
California (CA)	242	171 / 36 / 35	0.0173/ 0.0195/ 0.0214	1521.61 ± 271.42	6.48 ± 3.42
Hainan (HN)	392	310 / 42 / 40	0.0225/ 0.0305/ 0.0354	120.94 ± 77.37	2.29 ± 1.97

3.1. Forest disturbance mapping models

3.1.1. From SAM to SAMSR: A novel architecture for SAR-based forest disturbance mapping

The SAM architecture: The SAM included three components, the image encoder, prompt encoder, and mask decoder (Kirillov et al., 2023). The image encoder used a pre-trained Vision Transformer (ViT) to generate image embeddings. The prompt encoder handled both sparse (points, boxes, text) and dense (masks) prompts. The mask decoder, using modified Transformer blocks, applied self- and cross-attention to update image and prompt embeddings. After two decoder layers, the image embeddings were upsampled and passed through an MLP to a dynamic linear classifier, outputting the foreground mask probability for each position.

The SAMSR architecture: The SAMSR retained SAM's prompt encoder and mask decoder, adding a parallel CNN branch to the image encoder. The CNN and ViT branches established global dependencies via a Cross-Branch Attention (CBA) module, producing the final image feature embedding. In the ViT branch, added a position adapter and five feature adapters fine-tune position embeddings and aligned input features with the pre-trained ViT encoder. The first feature adapter adjusted input features to match the ViT encoder's distribution, while the remaining adapters refined the pre-trained encoder through residual connections to better suit the specific task.

The SAMSR specific details: As shown in Fig. 2, the input image was first preprocessed and tokenized, then fed into both the ViT and CNN branches. In the CNN branch, the image passed through a single convolutional block, followed by three convolutional pooling blocks, and then the same convolutional block was repeated four times. In the ViT branch, the image undergoes overlapped patch embedding and position embedding (patch size: 16 × 16), followed by optimization through the feature adapter and position adapter modules. The feature adapter consisted of two linear layers with an activation layer in between, followed by a residual connection, allowing the ViT branch to better adapt to the SAR image feature distribution. The position adapter was a convolutional layer that adjusted position embeddings to match the resolution of the embedding sequence, first by downsampled with max pooling (stride: 2, kernel size: 2) and then refining with a 3 × 3 convolution.

The embeddings were processed through the window transformer, where attention was computed only between patches within a 14 × 14 window, reducing computational complexity (Li et al., 2022a, 2022b). The embeddings then passed through multiple ViT transformer blocks (using the pre-trained ViT-B model with 12 layers of transformers) and interacted with every global transformer via the CBA module, injecting local CNN features into the ViT branch. The CBA module used CNN embeddings as key and value, while ViT embeddings serve as query. After fine-tuning each global transformer layer with the feature adapter, the output was combined with the feedforward network via a residual connection. The final image embeddings from both the ViT and CNN branches were summed and input to the mask decoder. Simultaneously, a point prompt was randomly selected from the label mask area, and a point prompt embedding was generated by the prompt encoder. The mask decoder combined embeddings from the ViT branch, CNN branch, and prompt, outputting the image's features. The image features were upsampled through convolution and multiplied by the query embedding to generate the predicted result. This was then interpolated back to the original resolution and compared with the label masks to calculate the loss, which was used for backpropagation to train the model.

Training Strategies: SAMSR initialized the parameters from SAM, and the pre-trained model used the initial weights of SAM's ViT-B

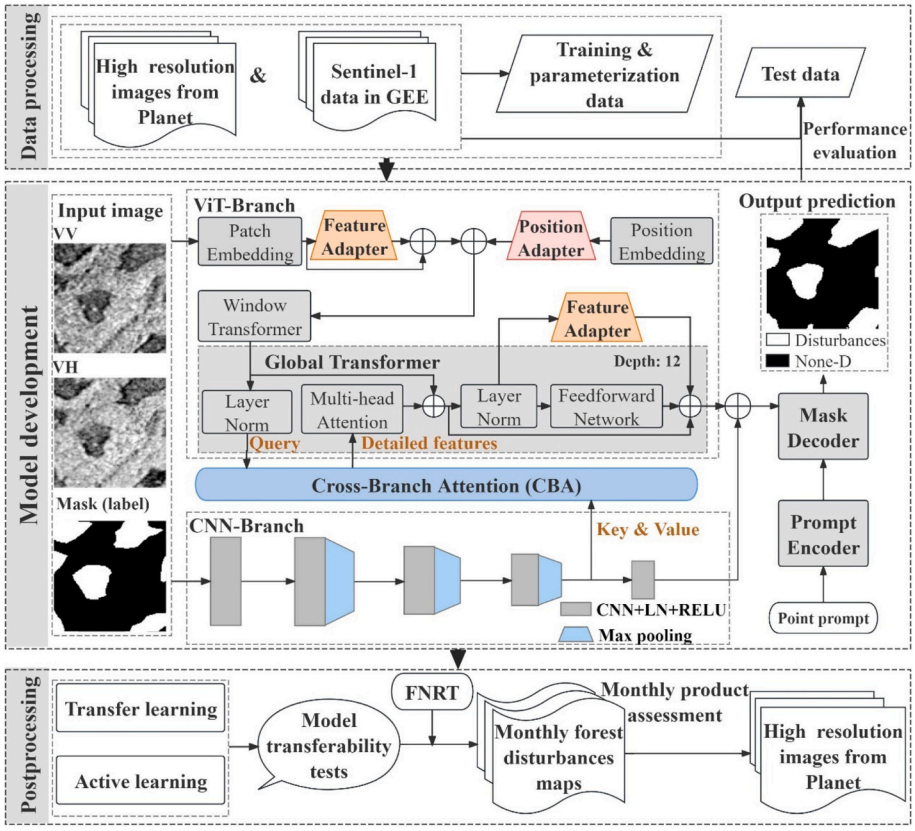


Fig. 2. The workflow of the SAMSR model.

(Kirillov et al., 2023). The number of samples used for training the model in the four areas was 198, 495, 242 and 392, respectively, and the specific details were shown in Table 1. The training process was supervised by a combined loss function, including dice loss and binary cross-entropy loss. The model was implemented within the PyTorch framework, we used Adam as the optimizer with an initial learning rate of 0.0001, a batch size of 8, and trained the model for 200 epochs. The input size was 128 × 128 pixels, with 2 bands (VV and VH), and the output size was 128 × 128 × 1, where each pixel value represented the probability of forest disturbance. Pixels with a probability greater than 0.5 were classified as disturbance.

3.1.2. Baseline models

We employed the U-Net model (Zhao et al., 2022) and the DeepLabv3+ model (Chen et al., 2018; Ma et al., 2024a, 2024b) as baselines to compare with the SAMSR. In the U-Net's 'downsampling' section, pooling reduced spatial dimensions, and we set up four convolutional layers where the channels expand from 8 to 128 to capture complex features. The 'upsampling' section then upscaled these features, merging them with corresponding downsampling features via skip connections to maintain spatial details. We used ReLU activation in all convolutional layers except the final one, where a sigmoid function outputs values between 0 and 1. The U-Net was trained using a combination of binary cross-entropy loss and dice loss, with the Adam optimizer set at an initial learning rate of 0.0001.

DeepLabV3+ was an advanced image semantic segmentation model that used MobileNetV2 (Sandler et al., 2018) as its backbone network. MobileNetV2 employed an efficient inverted residual structure and depthwise separable convolutions, which significantly reduce computational complexity while maintaining high feature representation capability. To better recover image boundaries and fine details, DeepLabV3+ introduced a decoder module, which combined low-level features from the encoder with high-level features through upsampled and skip connections, refining the segmentation results. Additionally, the model integrated atrous spatial pyramid pooling (ASPP) in the decoder, which extracted multi-scale features using convolutions with different dilation rates, further improving its segmentation performance in complex scenes. The loss function and hyperparameters used during model training were consistent with those in SAMSR.

3.2. Accuracy assessments

The performance of the classification models (SAMSR, U-Net, DeepLabv3+ and SAM) was evaluated using an independent test data. To assess model accuracy, we calculated several metrics, including Overall Accuracy (OA), User's Accuracy (UA), Producer's Accuracy (PA), F1 Score, and Intersection over Union (IoU) as defined in formulas (1)–(5). The F1 Score, representing the geometric mean of UA and PA, provides a more precise measure of model performance compared to OA. IoU quantifies the ratio of the intersection to the union area between predicted and actual results.

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$UA = \frac{TP}{TP + FP} \tag{2}$$

$$PA = \frac{TP}{TP + FN} \tag{3}$$

$$F1 = \frac{2*UA*PA}{UA + PA} = \frac{2TP}{2TP + FP + FN} \tag{4}$$

$$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union} = \frac{TP}{TP + FP + FN} \tag{5}$$

where TP, TN, FP, FN represent true positive, true negative, false posi-

tive, and false negative, respectively.

To evaluate the accuracy of the final monthly forest disturbance products, we randomly selected 100 quadrats (1280 × 1280 m) from the produced monthly products in each study area, evenly distributed across time and space during the study interval and area. These quadrats were independently assessed using Planet imagery. Two interpreters manually outlined disturbance and non-disturbance areas by synthesizing reference images from the target and months before and after. An independent analyst then compared these manually drawn reference maps with the final monthly disturbance maps to calculate the accuracy of the final monthly disturbance products.

"Sigmoid Accuracy" (Bullock et al., 2022) assessment framework was used to assess the spatiotemporal accuracy of our monthly disturbance products. This framework focused on the relationship between timeliness and accuracy, with key indicators including the "Initial Delay" and the "Level Off Point", which were derived from the "F1 score" curve as a function of "lag". The "Initial Delay" was identified as the point where the F1 begins to rise, while the "Level Off Point" was determined based on the saturation point of the F1. The "lag" started from 0, indicating that the detection month was consistent with the reference data, and 1 indicated that the detection month was one month later than the reference data, and so on. This approach provides a comprehensive assessment of the temporal and spatial accuracy of our monthly disturbance mapping results.

3.3. Evaluating the SAMSR'S transferability

To evaluate the transferability of our proposed framework across four study areas, we conducted the following model transfer experiments: (1) We trained the SAMSR model separately for each area (Rondônia in Brazil, Guangxi in China, California in the USA, and Hainan in China) using local samples, starting with 10 input objects and increasing the training data until performance plateaued (measured by IoU). (2) We then applied each saturated local model to the remaining three study areas, progressively adding local samples until saturation. (3) Finally, we applied integrated models trained on three areas to the fourth area, incrementally increasing samples.

Gradually increasing the training data allowed us to assess how performance improves with sample size and determine the minimum data required for stable results. Comparing the accuracy of local and transferred models at different sample sizes helped us understand the relationship between data size and performance, providing a more effective evaluation of the SAMSR model's transferability. In testing the model's transferability, we employed fine-tuning-based transfer learning approach. According to the transfer learning outlined by Ma et al. (2024a, 2024b), which categorizes transfer learning into five distinct types with fine-tuning being the most common. We began by initializing the network weights with pre-trained model developed using Sentinel-1 SAR data from source area. Then, all layers of the network were retrained using labeled samples from target area, which allowed the model to fully adjust both low- and high-level feature detectors to better capture the specific patterns and nuances of the target area. This comprehensive fine-tuning enabled the model to adapt its feature representations to the unique spatial characteristics of the target area while leveraging the generalized features learned from the source domain.

3.4. The effectiveness of active learning

When applying transfer learning models, augmenting application scenarios with new local samples is crucial. This paper employed an improved Greedy Sampling (iGS) method for selecting representative samples, which was shown to outperform traditional active learning algorithms, by achieving comparable performance with fewer samples while conserving computational resources (Wan et al., 2023). The iGS algorithm leveraged entropy to estimate the representativeness of selected samples, which is an important concept in information theory that measures the randomness or uncertainty of information (Shannon, 1948). In image processing, entropy measures an image's complexity and information content. High entropy indicates greater complexity and more information, providing a quantitative method for analysis. The formula is as follows:

$$H(X) = -\sum_{i=1}^{n} p(x_i) log_2 p(x_i) \quad (6)$$

where n is the number of different states that the random pixel value X can take, and $p(x_i)$ is the probability of X taking the value x_i.

The iGS process includes: (1) GS for input (GSx), selecting samples based on proximity to the centroid. (2) GS for output (GSy), selecting samples based on output value diversity. (3) iGS, combining input and output diversity to select samples. Iterations continue until the maximum sample number (K) is reached. To compare iGS with random sampling, nine sets of sample gradient experiments were conducted.

To demonstrate the effectiveness of active learning, T-SNE (Van der Maaten and Hinton, 2008) was used to visually show the features and distribution of selected samples. Here, random and active samples were plotted for each study area with a sample size of 30. Additionally, the clustering metric F-score of the samples was calculated to evaluate the impact of the active learning selection sample. Moreover, this study plotted the distribution histogram of the samples' VH values as well as the density plot and information entropy to further assess the performance of active learning.

3.5. Monthly forest disturbance map production and analysis

To quantitatively assess the dynamic of large-scale mapping results, it is crucial to integrate temporal change analysis into the methodology. This study utilized the monitoring window in Fusion Near Real-Time (FNRT) approach to continuously monitor the change of time series (Tang et al., 2023a, 2023b). The FNRT monitoring window based on the results from the SAMSR to create dense time series, on which the method operated to monitor disturbances and detect dates. Although the SAMSR had segmented the disturbance patches, noise in the data may cause the model to incorrectly segment an image, leading to errors. To reduce noise while ensuring timely detection, the FNRT monitoring window minimized noise effects by "confirm" changes through multiple successive detections. Model parameters, the monitoring window size (n) and consecutive changes (k), determined the method's sensitivity. The method identified disturbances when k consecutive changes occurred within the monitoring window, marking the first change's date. After testing various parameters, the final disturbance map was generated using $n = 5$ and $k = 3$. Fig. S6 illustrates how the FNRT monitoring window to continuously monitor the results of the SAMSR. Using the optimal mapping model, monthly disturbance regions were mapped for four study areas from 2016 to 2023 (including 96 consecutive months). First, non-forest areas were excluded using the Hansen GFC product. Then, the FNRT algorithm assigned year-month labels to each disturbance area. A threshold of 900 m^2 was applied to remove small patches, and only areas overlapping with Hansen GFC changes by more than 50 % were included in the final maps.

4. Results

4.1. Accuracy assessments of forest disturbance mapping models

Using randomly selected disturbance patches, we plotted the time series boxplots of SAR backscattering coefficients (VV and VH boxplots of nine months before and after forest disturbance) for the four study areas (Fig. S1). The VH coefficients showed significant deviations between stable forests and disturbances across all study areas (Ron, GX, CA, and HN). In contrast, the VV values (Fig. S1a, S1c, S1e, and S1g) displayed less pronounced variations, although disturbances consistently showed slightly lower VV values compared to stable forests. These

findings highlight the utility of VH coefficients as a critical indicator in detecting forest disturbance events across diverse geographical environments, making it a valuable tool for monitoring forest health using SAR data.

To rigorously examine the specific role played by each module and assess their true contribution to the model's performance, we conducted ablation study by sequentially introducing the four modules from SAMSR—CNN branch, CBA, feature adapter, and position adapter—into the SAM, with the results shown in Table S1. Each module can effectively improve the segmentation performance and generalization ability of SAM on SAR images. In particular, the addition of the feature adapter module led to significant increase in F1 score over the SAM (with local sample fine-tuning), with improvements of 7 %, 14 %, 7 %, and 10 % across four study areas. Furthermore, simply fusing the output of the CNN and ViT branches was not the best approach, introducing the CBA module further facilitated the integration of local features. By coupling these four components, SAMSR achieved the best segmentation performance and generalization results.

Table S2 showed the results of the original SAM (without local sample fine-tuning) for direct predictions. The results indicated that the original SAM performed the worst, with an F1 score of only 0.09 on the comprehensive dataset. After fine-tuning with samples (SAM), the F1 score quickly increased to 0.68, and after improvements, the F1 score to 0.78.

Based on independent test data, the SAMSR outperformed the U-Net, DeepLabv3+ and SAM in all scenarios, with an overall accuracy ranging from 86 % to 94 % (the combined model for the four study areas achieved 88 %). The user's and producer's accuracies were 69 %–83 % and 66 %–93 %, respectively (for the combined model of the four study areas, the user's and producer's accuracies were 79 % and 77 %, respectively) (Table 2). The SAMSR's F1 score and IoU ranged from 0.67 to 0.88 and 0.50 to 0.78, respectively (the combined model for the four study areas achieved an F1 score of 0.78 and an IoU of 0.64). The combined model based on DeepLabv3+ for the four study areas achieved an overall accuracy of 88 %, with F1 score of 0.77 and IoU of 0.63, respectively. The U-Net's F1 score and IoU were 0.75 and 0.62, respectively. The SAM had the lowest accuracy, with F1 of 0.68.

Comparing the spatial mapping results with the reference and baseline predictions, the SAMSR showed the greatest similarity to the reference labels in spatial location and boundary consistency (Fig. 3). The U-Net model produced more false negatives than SAMSR. In contrast, SAM's predictions were the least accurate overall.

Table 2
Values of the evaluation results of the independent test plots.

Study areas	Models	OA	UA	PA	F1	IoU
Ron	U-Net	0.88	0.85	0.89	0.87	0.76
	DeepLabv3+	0.89	0.82	0.92	0.87	0.77
	SAM	0.84	0.78	0.81	0.79	0.66
	SAMSR	**0.90**	**0.83**	**0.93**	**0.88**	**0.78**
GX	U-Net	0.83	0.82	0.69	0.75	0.60
	DeepLabv3+	0.84	0.81	0.78	0.79	0.64
	SAM	0.76	0.70	0.60	0.65	0.47
	SAMSR	**0.86**	**0.82**	**0.77**	**0.79**	**0.66**
CA	U-Net	0.93	0.87	0.70	0.77	0.63
	DeepLabv3+	0.94	0.80	0.85	0.82	0.67
	SAM	0.91	0.74	0.72	0.73	0.56
	SAMSR	**0.94**	**0.79**	**0.84**	**0.81**	**0.68**
HN	U-Net	0.87	0.71	0.55	0.62	0.45
	DeepLabv3+	0.86	0.63	0.61	0.62	0.45
	SAM	0.83	0.56	0.51	0.53	0.36
	SAMSR	**0.87**	**0.69**	**0.66**	**0.67**	**0.50**
All data	U-Net	0.88	0.78	0.73	0.75	0.62
	DeepLabv3+	0.88	0.79	0.76	0.77	0.63
	SAM	0.83	0.71	0.66	0.68	0.52
	SAMSR	**0.88**	**0.79**	**0.77**	**0.78**	**0.64**

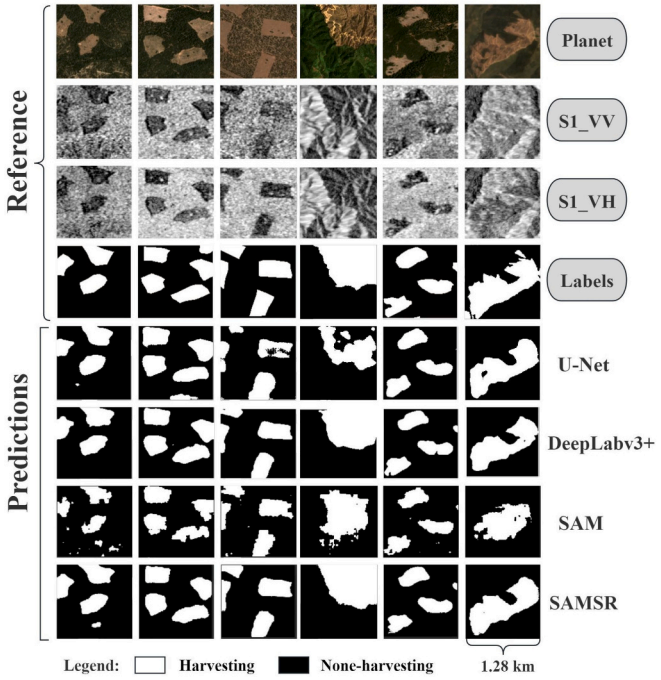


Fig. 3. Spatial display of the reference and prediction results from the U-Net, DeepLabv3+, SAM and SAMSR models. The first row presents Planet high-resolution images, followed by VV and VH of Sentinel-1 in the second and third rows. The fourth row contains manually drawn labels, the fifth, sixth, seventh and eighth rows show the prediction results of the U-Net, DeepLabv3+, SAM and SAMSR models, respectively.

4.2. SAMSR transferability assessment

Results showed that more complex patches require more samples to fully train the model, whereas simpler patches require fewer (Table S3). The PAR in Hainan was significantly higher than that of the other areas, indicating greater mapping challenges in this region (Table 1). The IoU in Hainan reached saturation at around 150 local training patches, while in Ron, with the lowest PAR, it saturates at 100 patches, and reached 0.7 IoU with just 40 patches. Similar trends were observed in F1 and OA across all areas (Table S4).

The complexity of patches also affected the model's transferability. Applying a fully trained model from one area to another without fine-tuning did not yield high accuracy. However, fine-tuning with sparse local patches improved performance significantly. This spatial transferability may be influenced by the variations in the complexity of the disturbance patches shapes (Table S3). For example, a model pre-trained in Ron, when applied to Hainan, initially achieved an IoU of 0.31. Retraining with 10 local patches from Hainan increased the IoU to 0.56, and the IoU saturated at 150 patches. Conversely, applying a Hainan-trained model to Ron reached 0.71 in IoU with just 10 samples, while the IoU saturated at 40. These findings suggest that areas with more complex patches (higher PAR) require more local samples for effective model transfer. To further explore transferability, models trained on three areas were applied to the fourth, with gradually increasing sample numbers. This approach generally yielded better accuracy compared to single-area transfers (Table S3), with final accuracies consistently higher or similar to other methods. Maximum F1 and IoU values for each area were summarized in Table 3.

4.3. Effectiveness of active learning for SAMSR accuracy and efficiency

This study demonstrated that active learning consistently outperforms random sampling in improving model accuracy across study areas. Specifically, as shown in Fig. 4, Fig. S2–S5, and Table S5, the

Table 3
The maximum F1 and IoU values for the transferability testing for study areas. The values with different samples are shown in Table S1 of the supplementary material.

Study areas	Maximum F1 / IoU for the models				
Ron	Local	GX	CA	HN	Multi-region
	0.87/0.77	0.87/0.77	0.86/0.77	0.86/0.76	0.87/0.77
GX	Local	Ron	CA	HN	Multi-region
	0.77/0.62	0.75/0.60	0.77/0.62	0.78/0.64	0.79/0.65
CA	Local	Ron	GX	HN	Multi-region
	0.81/0.68	0.83/0.72	0.81/0.68	0.82/0.71	0.84/0.74
HN	Local	Ron	GX	CA	Multi-region
	0.73/0.58	0.75/0.60	0.75/0.60	0.74/0.58	0.75/0.60

active learning method achieved higher accuracy, particularly with fewer samples, reducing annotation costs. Additionally, the Fig. 4 highlighted the transferability of active learning in multi-region models, where it reached saturation with fewer samples while maintaining superior accuracy.

Active learning technique can meet the saturation accuracy of traditional random selection with about 20 %–70 % less training samples. Specifically, the GX model transferred to Ron achieved an IoU of 0.76 with 30 samples using active learning, comparable to or surpassing random sampling with 100 samples (Fig. S2b). In Ron, active learning consistently outperformed random methods under all sample conditions, with accuracy saturating around 40 samples (Fig. S2a), a trend also observed in GX models. The difference in accuracy between the two methods was more pronounced in GX models, with active learning achieving higher final saturation accuracy (Fig. S3). While the difference between the two methods in CA and HN was smaller, active learning still required fewer samples to reach saturation compared to random methods (Fig. S4 and S5). These findings highlight the advantage of active learning, where fewer samples can achieve or exceed the saturation effect of random sampling.

The T-SNE plots (Fig. 5, left) and VH value distribution plots (Fig. 5, middle and right) demonstrated that the samples selected through active learning were more evenly distributed in the feature space, indicated that they covered the underlying data distribution more extensively and were thus more representative. The higher entropy values observed across all study areas further support this point (Fig. 5, right), suggested that the active samples were more effective in capturing the diversity and key features of the data. The higher F-scores also suggested that these samples were more effective in capturing the intrinsic structure of the data (Fig. 5, left), making them a more valuable choice for training models. In contrast, the random samples exhibited a narrower distribution and lower entropy, indicated that they captured less of the dataset's complexity. Although the entropy values for random samples were higher than those for the overall sample in most cases, capturing some diversity, they often failed to fully represent the complexity of the data, particularly in certain cases, such as at GX (Fig. 5b). This further underscore the effectiveness and stability of the active learning approach. Overall, these visualizations and metrics indicated that the samples selected through active learning were more representative, exhibited a broader distribution, and contained richer information compared to random samples and the overall dataset.

4.4. Monthly forest disturbance mapping from 2016 to 2023

To further evaluate the spatiotemporal accuracy of the monthly disturbance products, we analyzed the relationship between lag and F1-scores (Fig. 6). At an initial delay (lag = 0), F1-scores for Ron, GX, CA, and HN were 0.49, 0.49, 0.72, and 0.60, respectively. As the lag increased, the F1-scores gradually increased and saturated at 0.83, 0.61, 0.89 and 0.63, respectively. At this point, the corresponding lags were four months, three months, three months, and one month, respectively. The results revealed varying spatiotemporal accuracies across the study areas. The CA and Ron achieved higher F1-scores, indicating that the proposed method performed well in these study areas. In contrast, GX and HN exhibited relatively lower F1-scores, likely because areas had more fragmented disturbance patches (mean perimeter-to-area of 0.0300 and 0.0305, respectively, compared with 0.0195 and 0.0122 in CA and Ron), which posed greater challenges for the model. These findings validated the overall effectiveness of SAMSR in mapping monthly disturbance while highlighting the influence of disturbance patch complexity on model performance.

Utilizing the optimal mapping framework identified earlier (the SAMSR), we mapped and analyzed monthly disturbance maps for the four areas from 2016 to 2023. During the study period, the disturbed areas of Ron were approximately 819.81 km^2, equivalent to 7.46 % of the study area. The most significant losses occurred in 2017, approximately 170.28 km^2, with a peak in August (approximately 42.26 km^2, Fig. 7a). In GX, the total disturbed areas were approximately 121.03 km^2, accounting for about 1.10 % of the study area (Fig. 7b). For CA, the

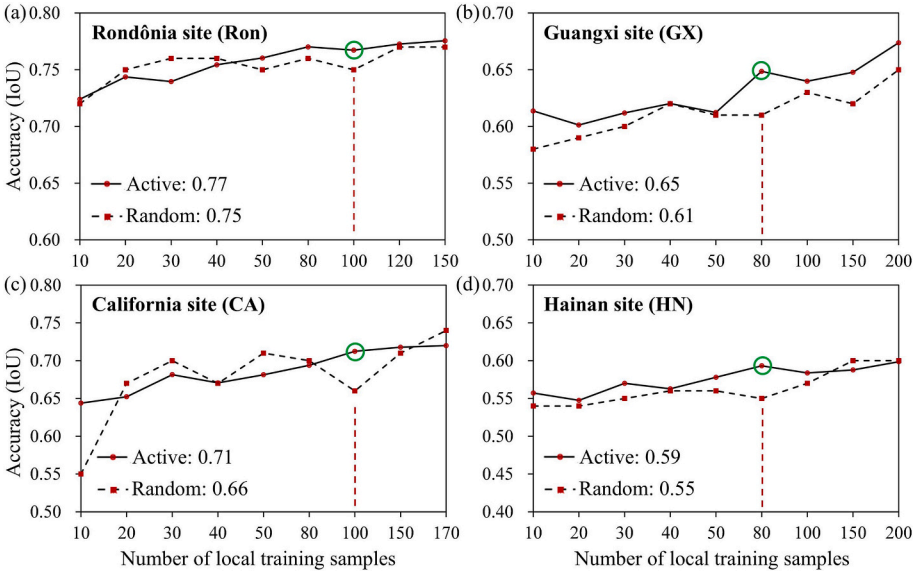


Fig. 4. The IoU values of random and active methods under different samples for the Ron (a), GX (b), CA (c), and HN (d) areas. The figure represents a multi-region synthesis model for three additional areas. For the single area model, see Fig. S2-S5 in the supplementary material.

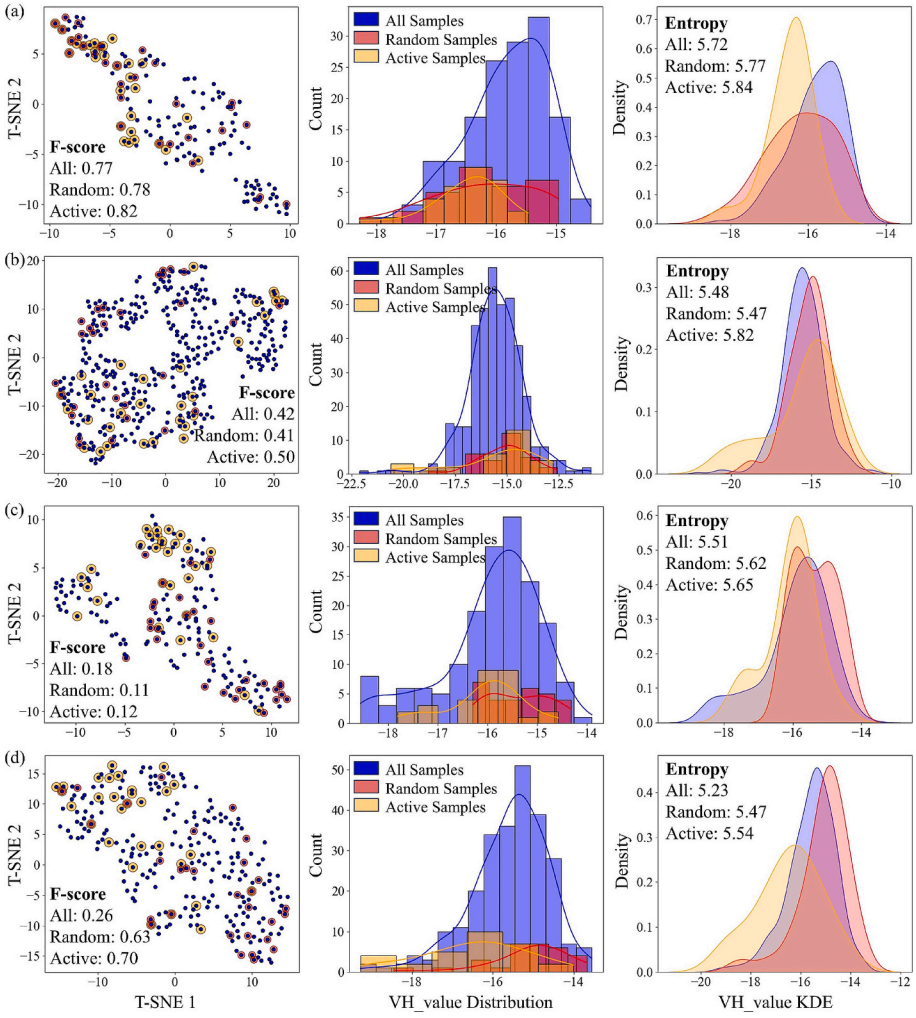


Fig. 5. T-SNE visualization and clustering metric F-score (left), VH value distribution histogram (middle), and VH value density distribution and information entropy (right) for all samples, random samples, and active samples in (a) Ron, (b) GX, (c) CA, and (d) HN.

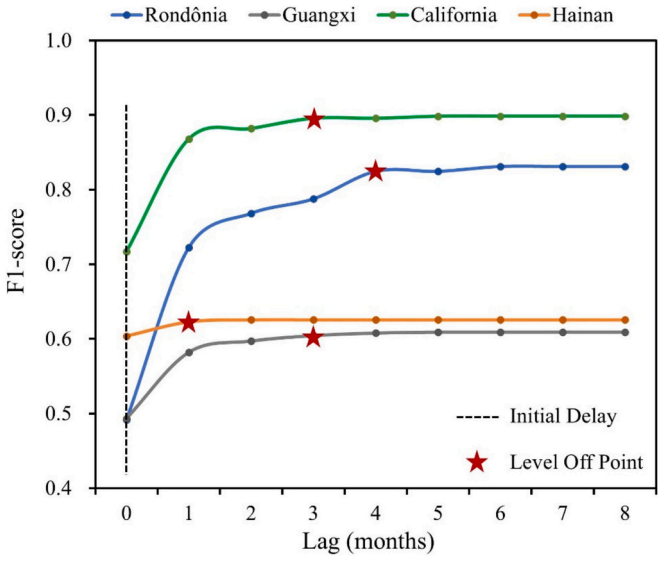


Fig. 6. The spatiotemporal accuracy illustrated by the lag / accuracy curve of monthly forest disturbance mapping. F1 gradually increases with higher lag and reaches the Level Off Point (stars).

total disturbed areas were approximately 493.06 km^2, accounting for about 2.98 % of the study area (Fig. 7c). Disturbance was most prevalent in 2021 (approximately 276.52 km^2, and 23.04 km^2 per month), followed by 2022 and 2020. In HN, the total disturbed areas were approximately 239.58 km^2, accounting for about 2.12 % of the study area (Fig. 7d). In Ron, the disturbances occurred mainly during the dry season, especially at the end of the dry season (August–October). The interannual variation of disturbance in CA was large, which showed the influence of single disturbance event. In GX and HN, the disturbance was more evenly distributed throughout the year.

5. Discussion

In this study, we proposed a novel framework that leverages the SAM, a vision foundation model, and Sentinel-1 SAR data to map forest disturbance at fine spatial-temporal scales (Fig. 2). Results showed that SAMSR (IoU 0.50–0.78; F1 0.67–0.88) outperformed the baseline SAM, U-Net and DeepLabv3+ methods for about 23.08 %, 3.23 % and 1.59 % in IoU, respectively, with high transferability across four distinct forest disturbance hotspots (Rondônia in Brazil, Guangxi in China, California in the USA, and Hainan in China) (Table 2, Fig. 3). The best model transfer strategy across-areas based on transfer learning was the combined model trained on multi-region samples, with IoU from 0.60 to 0.77, F1 from 0.75 to 0.87 (Table 3, Table S3, Table S4). Furthermore, active learning significantly enhanced the model's efficiency and

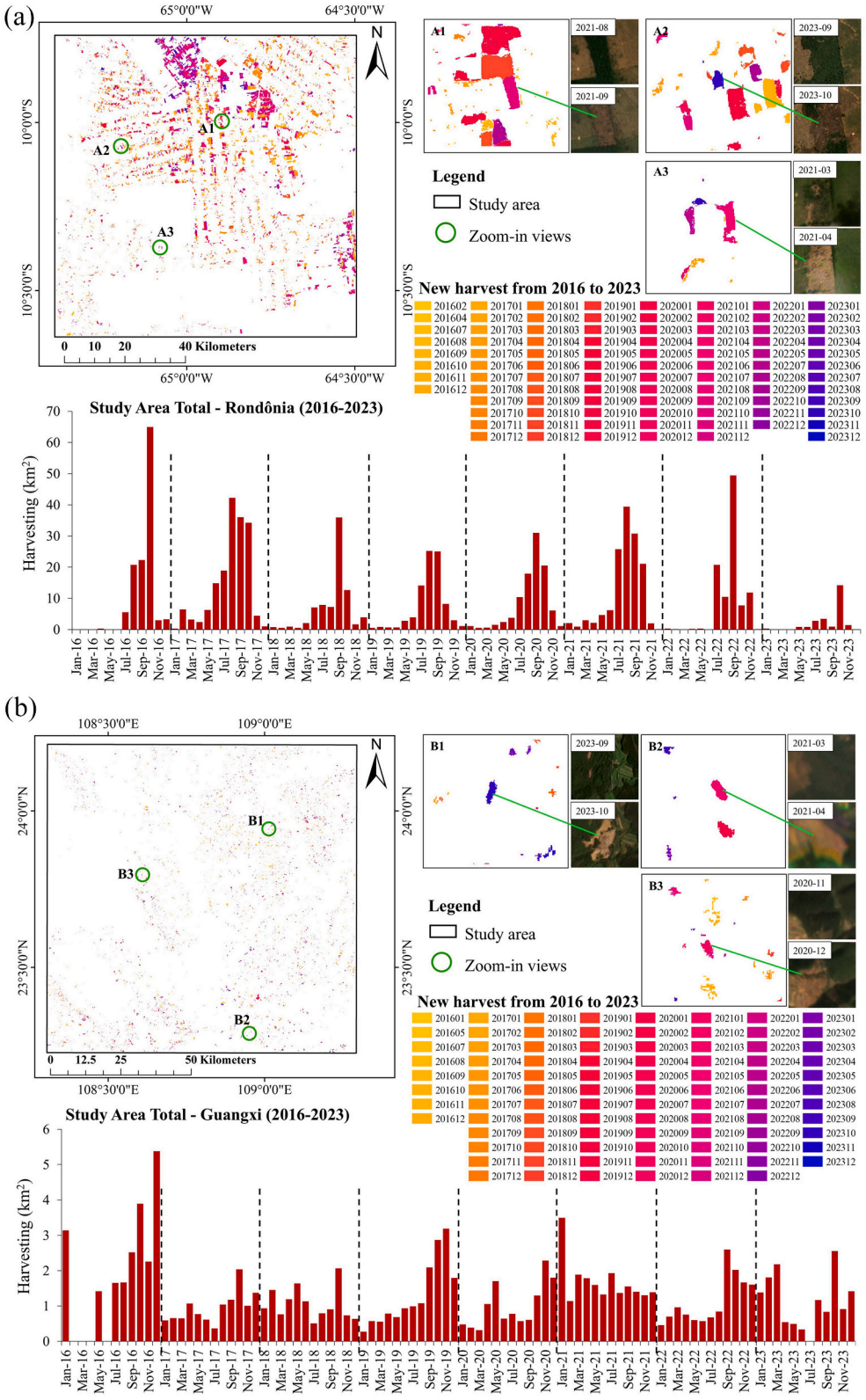


Fig. 7. (a). Monthly disturbance results for the study area in Rondônia from 2016 to 2023. Three green circle areas are shown with zoom-in views. Adjacent to these zoomed-in views are images from Planet, depicting the months before and after disturbance. The bar chart is the monthly statistical chart of disturbed forest area.
(b). Monthly disturbance results for the study area in Guangxi from 2016 to 2023. Select the central area of Guangxi for mapping and analysis.
(c). Monthly disturbance results for the study area in California from 2016 to 2023.
(d). Monthly disturbance results for the study area in Hainan from 2016 to 2023. Select the central area of Hainan for mapping and analysis.

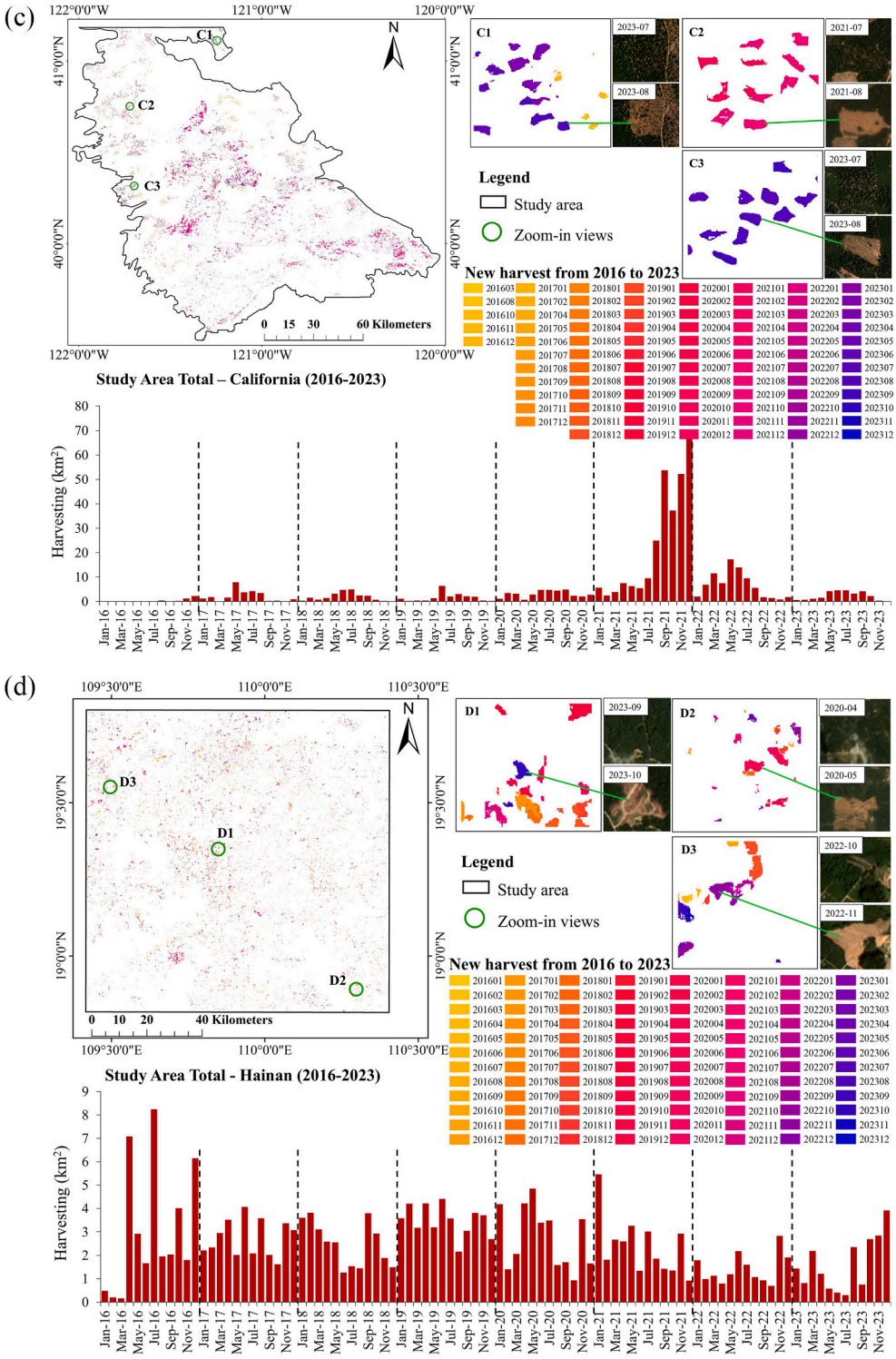


Fig. 7. (*continued*).

performance by achieving or exceeding the saturation accuracy of traditional random selection methods with 20 %–70 % less samples (Table S5, Fig. 4, Fig. S2-S5).

5.1. *Performance of the SAMSR in disturbance mapping based on Sentinel-1 data*

The proposed SAMSR significantly enhanced the ability to detect forest disturbance in SAR images. Compared with the SAM, U-Net and DeepLabv3+ models, the IoU improved by about 23.08 %, 3.23 % and 1.59 %, respectively (Table 2). The SAM has demonstrated excellent performance in various scene segmentation tasks, such as buildings, roads, and crop monitoring, based on optical images (Song et al., 2024; Zhang et al., 2024). However, integrating SAM and S1 SAR for forest disturbance mapping involves unique challenges due to the differences between optical and SAR imagery. Optical data, while offering rich spectral and textural information, is highly dependent on external factors such as weather conditions and seasonal changes (Alshehri et al.,

2024). This issue makes it difficult to obtain consistent, high-quality images for accurate disturbance detection, especially in tropical regions where cloud coverage is frequent and persistent. Due to the complex and dynamic nature of forest environments, applying SAM to forest disturbance mapping using SAR data is vital but also more challenging because of the presence of speckle noise and the variation in the size, shape, and intensity of disturbance patches (Tsokas et al., 2022). Therefore, we modified the model by incorporating the CNN branch and cross-branch attention, which enhanced the extraction of local features and the integration of global information, significantly improving the detection of forest disturbance (Table 2, Table S2, Fig. 3). The CNN offered significant advantages in extracting small targets and recognizing fine details, while the cross-branch attention established information flow between global and local features, further strengthening the synergy between them. This approach has been applied in extracting features from medical images, particularly ultrasound images, successfully capturing small targets and weak boundary objects (Lin et al., 2023; Tu et al., 2024), while our study serves as the first investigation for adapting SAM to detect forest disturbance from S1 SAR images. Our modifications further improved SAM's generalization ability in SAR tasks and enhanced its adaptability in complex image segmentation tasks. Traditional methods often struggle to capture the complex details of disturbance patches (Kattenborn et al., 2021), while the modified SAM overcame these challenges by integrating global and local features, enabling accurate detection of subtle disturbance. By effectively applying SAM to SAR data, the proposed approach provides new possibilities for large-scale, accurate monitoring of global forest disturbance. However, despite these improvements, there is still potential for further optimization. Future research could explore effective methods for integrating and analyzing multi-modal data within the SAM framework, which would enhance the model's adaptability and reliability across a wider range of data scenarios (Li et al., 2023; Tang et al., 2023a, 2023b).

The superior performance of SAMSR could be attributed to its fine-tuning, which leveraged pre-trained models, a strategy that was both cost-effective and practical in real-world applications. Training large models from scratch requires significant computational resources and large datasets (Ahmed et al., 2023; Wang et al., 2021a, 2021b), which many institutions cannot afford. In contrast, fine-tuning pre-trained models like SAM enables high performance with fewer training samples, making it a valuable approach. Thus, the comparison with a small model like U-Net is not only about model capacity, but also about the practical feasibility of using a large pre-trained model. Furthermore, to better understand the detection capabilities and generalization ability of the SAMSR model, we compared the detection results with the GLAD forest disturbance dataset (Hansen et al., 2016; Pickens et al., 2020; Reiche et al., 2021). For each study area, we randomly selected 20 quadrats (1280 × 1280 m) and calculated the accuracy separately according to the year. Since the GLAD dataset did not include the CA area, we compared it with the other three areas. According to the Table S6, in the Ron area, the GLAD dataset's accuracy was slightly higher than our results (mean GLAD-F1 score was 0.07 higher), but for the GX and HN areas, the advantage of our results was very clear (mean SAMSR-F1 score was 0.55 and 0.67 higher, respectively). While GLAD combined data from three sources (Landsat, Sentinel-2, Sentinel-1), our method, using only Sentinel-1 data, achieved comparable accuracy, highlighting its advantage. Moreover, we found that the GLAD dataset missed many detections of GX and HN areas (Fig. S7), compared with other areas, the GX and HN showed more complex landscape patterns, indicated that our method has greater advantages in detecting small-scale disturbance and complex landscape patterns. One possible reason for this was that the samples of the GLAD-L method were selected in wet forests, which may have limitations when applied to pantropical regions (Hansen et al., 2016). Despite the high potential of SAR data for consistent mapping in cloud-prone regions (Mullissa et al., 2023; Reiche et al., 2021; Zhao et al., 2022), the presence of speckle noise and the lack of spectral information limits its capabilities (Guo et al., 2022), and optical data could provide complementary spectral information. Some techniques such as multi-temporal synthesis, gap-filling, and multi-sensor data fusion (Qiu et al., 2023; Wang et al., 2021a, 2021b; Zhou et al., 2021), could contribute to the fusion of SAR and optical data for improved accuracy and robustness in challenging environments, which are beyond the scope of this study. For future studies, we could explore the hybrid architecture of SAR and optical image fusion, integrating complementary information from multiple sources to further improve SAR monitoring capabilities (Liu et al., 2024; McGregor et al., 2024; Reichstein et al., 2019). These efforts would help overcome the inherent limitations of SAR images and extend the applicability of models like SAMSR to a broader range of remote sensing tasks.

5.2. Application and advantages of transfer learning in forest disturbance mapping

This study demonstrated the usefulness of transfer learning in forest disturbance mapping: the transfer learning technique reduced the need for extensive labeled data while enhancing the model's generalization, maintaining efficiency across various environments. The model achieves high accuracy in different areas with minimal local training data, making it valuable for large-scale, resource-limited forest monitoring (Table 3, Table S3, Table S4). Transfer learning significantly improved the accuracy of the model, especially the combined model with multi-region achieved the best transfer performance (Table 3). This improvement may be due to the similarity within disturbance datasets, allowing the model to leverage learned patterns effectively (Ehrig et al., 2024; Ma et al., 2024a, 2024b). However, in CA, without local samples, the model pre-trained in HN (IoU = 0.55) outperformed the combined model (IoU = 0.49), likely due to the higher PAR and complexity of patches in CA and HN (Table 1). As more local data became available, the combined model's advantages became more apparent with the local sample 30 was added, the combined model (IoU = 0.70) exceeds the HN model (IoU = 0.69), and the final IoU was 4.2 % higher than the HN model.

In this study, we gradually increased the training data to assess how sample size affects model performance and transferability (Table S3-S4). By comparing local models with transferred models, we gained deeper insights into the SAMSR model's ability to transfer across regions and its transferability with varying sample sizes. This approach provided a more reliable understanding of the model's adaptability and its potential for broader application. The application of transfer learning should be grounded on a solid understanding of the underlying remote sensing mechanism. The success of our proposed model in transfer learning was largely due to its ability to capture landscape patterns, which were relatively stable and transferable across different regions, and thus enabled effective transfer (Zhao et al., 2022). However, the success of transfer learning is not solely dependent on the similarity between source and target domains, but also on the robustness of the model to adapt to different contexts (Weiss et al., 2016; Zhuang et al., 2020). SAMSR's architecture, designed to capture these landscape patterns, plays a crucial role in this adaptability.

5.3. Active learning techniques improve forest disturbance mapping efficiencies

The initial performance of model without adaptation was suboptimal due to the non-stationarity of sample distribution across study areas (Table S3–S4). This requires incorporating target domain samples to improve the model's cross-spatial generalization. Traditionally, this involves random sample selection, which can negatively impact model performance if selected ineffectively (Hamrouni et al., 2021). Active learning enhances training efficiency by selecting the most representative samples, minimizing the number of samples needed for domain adaptation while maintaining high recognition capability (Alajlan et al.,

2013; Crawford et al., 2013). When the comprehensive model was transferred to CA with the addition of 10 local samples, the model accuracy (IoU = 0.55) was much lower than that of the HN pre-trained model (IoU = 0.63). However, with the participation of the active learning method, the accuracy of the comprehensive model rapidly improved (IoU = 0.64) and approached the accuracy of the HN model (IoU = 0.66) (Table S5). In some cases, the number of additional samples needed by the random method was twice that of the active method. For example, when transferring the HN model to Ron, to achieve an IoU of 0.77, the active method required 40 samples, whereas the random method required 100 samples. Additionally, when the same number of additional samples were added, the accuracy of the active method was significantly higher than that of the random method (Fig. 4, Fig. S2–S5, Table S5). These findings, consistent with those of Alajlan et al. (2013), validated that active learning was an effective approach to minimizing the number of samples needed for domain adaptation while maintaining high recognition capability. Similarly, in the mapping of poplar plantations in France, active learning improved overall accuracy by 5 % compared to random sampling, requiring only 50 % of the samples to achieve similar performance (Hamrouni et al., 2021).

Accurate sample labeling is crucial for large-scale forest disturbance mapping, but it also presents a significant challenge due to the complexities of labeling diverse and extensive areas. This study manually annotated disturbance masks using high-resolution Planet imagery, ensuring accurate labeling of subtle disturbances. However, this may introduce potential biases, as this process was susceptible to subjectivity and errors, which can lead to label noise (Peng et al., 2024). Nevertheless, most studies on remote sensing-based forest change monitoring assumed that manually annotated data accurately represented the underlying reality and were considered a widely accepted method in the field (Curtis et al., 2018; Senf and Seidl, 2021; Slagter et al., 2023). However, Planet imagery was costly, and its availability may be limited in remote or low-demand areas, hindering large-scale training. Fortunately, Planet has an official research program that allows researchers to apply for free access to their data for scientific purposes (https://www.planet.com/industries/education-and-research/). In addition, Norway's NICFI initiative provided free access to Planet's high-resolution data for the N30° ~ S30° tropical region, with updated ranging from biannual to monthly, enhancing the feasibility of tropical forest monitoring (Dalagnol et al., 2023). Furthermore, the reference data for validation can be substituted with other high-resolution imagery, such as Worldview or Google Earth images. Moreover, this further explained the necessity of our transfer and active learning to reduce our reliance on planet data and hand-annotated. Despite this, data availability and cost issues may persist for other regions in future studies. To address this issue, for future study, we will further strengthen the generality and transferability of the model, reduce the dependence on labels by exploring semi-supervised or unsupervised pre-training methods. Moreover, while spatial dimension plays a crucial role in forest disturbance detection (Hussain et al., 2013; Zhao et al., 2022), temporal information should not be overlooked, the fusion of spatial and temporal information could further improve detection accuracy. In future research, we will explore the 3D deep learning to effectively integrate both spatial and temporal dimensions for enhanced detection performance.

The monthly forest disturbance mapping framework based on the SAMSR offers promising practical applications, particularly in tropical regions with frequent cloud cover that limits optical data. Although there have been successful applications of Sentinel-1 for forest disturbance detection (Hethcoat et al., 2021; Reiche et al., 2018), these methods often lack testing for cross-regional transferability and the use of advanced techniques to further improve efficiency, which significantly limits the scalability of the models for large-scale applications. In contrast, SAMSR demonstrated significant advantages by addressing these gaps, incorporating both transfer learning and active learning techniques to enhance model efficiency and robustness across diverse areas. Overall, this study provides strong technical support for large-scale, across-areas forest disturbance mapping, offering significant scientific and practical value.

6. Conclusion

This study proposed and evaluated the SAMSR framework for mapping large-scale forest disturbance using Sentinel-1 images and an adapted SAM model. The proposed SAMSR demonstrated robust performances in four pan-tropical areas (IoU 0.50–0.78; F1 0.67–0.88). We further determined the optimal transfer strategies across-areas, and explored the potential of active learning to improve model efficiency. We found the combined model with multi-region achieved the best transfer performance and active learning further improved efficiency with about 20 %–70 % less training samples to reach saturation accuracy. This novel framework enables accurate and timely forest disturbance mapping at large-scales and supports forest management and protection by providing detailed insights into long-term forest dynamics.

CRediT authorship contribution statement

Yuping Tian: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Feng Zhao:** Writing – review & editing, Supervision, Conceptualization. **Ran Meng:** Writing – review & editing, Conceptualization. **Rui Sun:** Writing – review & editing, Software, Data curation. **Yuan Zhang:** Software, Investigation, Data curation. **Yanyan Shen:** Writing – review & editing, Software. **Bin Wang:** Writing – review & editing. **Jie Liu:** Writing – review & editing. **Mingze Li:** Writing – review & editing, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by National Natural Science Foundation of China (No. 42371395), National Key Research and Development Program of China (No. 2023YFD2201704), the Fundamental Research Funds for the Central Universities (No. 2572024BR33, No. 2572024AW01); Dr. Ran Meng was supported by National Natural Science Foundation of China (No. 42471362) and the Key Research and Development Program of Heilongjiang, China (grant No. 2022ZX01A25; No. JD2023GJ01).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2025.114775.

Data availability

Data will be made available on request.

References

Ahmed, S.F., Alam, M.S.B., Hassan, M., Rozbu, M.R., Ishtiak, T., Rafa, N., Mofijur, M., Shawkat Ali, A., Gandomi, A.H., 2023. Deep learning modelling techniques: current progress, applications, advantages, and challenges. Artif. Intell. Rev. 56, 13521–13617.

Alajlan, N., Pasolli, E., Melgani, F., Franzoso, A., 2013. Large-scale image classification using active learning. IEEE Geosci. Remote Sens. Lett. 11, 259–263.

Alshehri, M., Ouadou, A., Scott, G.J., 2024. Deep Transformer-Based Network Deforestation Detection in the Brazilian Amazon Using Sentinel-2 Imagery. IEEE Geosci. Remote Sens. Lett. 21.
Altarez, Armando Apan, Maraseni, T., 2023. Deep learning U-Net classification of Sentinel-1 and 2 fusions effectively demarcates tropical montane forest's deforestation. Remote Sens. Appl. Soc. Environ. 29, 100887.
Ballère, M., Bouvet, A., Mermoz, S., Le Toan, T., Koleck, T., Bedeau, C., André, M., Forestier, E., Frison, P.L., Lardeux, C., 2021. SAR data for tropical forest disturbance alerts in French Guiana: Benefit over optical imagery. Remote Sens. Environ. 252, 112159.
Brockerhoff, Luc Barbaro, Castagneyrol, Bastien, Forrester, David I., Gardiner, Barry, González-Olabarria, José Ramón, Lyver, Phil O'.B., Meurisse, Nicolas, Oxbrough, Anne, Taki, Hisatomo, Thompson, Ian D., Jactel, F.V.D.P.H., 2017. Forest biodiversity, ecosystem functioning and the provision of ecosystem services. Biodivers. Conserv. 26, 3005–3035.
Bullock, E.L., Healey, S.P., Yang, Z., Houborg, R., Gorelick, N., Tang, X., Andrianirina, C., 2022. Timeliness in forest change monitoring: A new assessment framework demonstrated using Sentinel-1 and a continuous change detection algorithm. Remote Sens. Environ. 276, 113043.
Cao, X., Yao, J., Xu, Z., Meng, D., 2020. Hyperspectral image classification with convolutional neural network and active learning. IEEE Trans. Geosci. Remote Sens. 58, 4604–4616.
Chen, L.C., Zhu, Y., Papandreou, G., Schroff, F., Adam, H., 2018. Encoder-decoder with atrous separable convolution for semantic image segmentation. In: Proc. European Conference on Computer Vision, pp. 801–818.
Chen, N., Tsendbazar, N.E., Hamunyela, E., Verbesselt, J., Herold, M., 2021. Sub-annual tropical forest disturbance monitoring using harmonized Landsat and Sentinel-2 data. Int. J. Appl. Earth Obs. Geoinf. 102, 102386.
Cortez, D., Soto, J., Román-Figueroa, C., Paneque, M., 2024. Monitoring loss and degradation of forests and shrubs in the North of Chile using Landsat time series data sets from 1998 to 2018. Remote Sens. Appl. Soc. Environ. 35, 101244.
Crawford, M.M., Tuia, D., Yang, H.L., 2013. Active learning: Any value for classification of remotely sensed data? In Proc. IEEE 101, 593–608.
Curtis, P.G., Slay, C.M., Harris, N.L., Tyukavina, A., Hansen, M.C., 2018. Classifying drivers of global forest loss. Science 361, 1108–1111.
Dalagnol, R., Wagner, F.H., Galvão, L.S., Braga, D., Osborn, F., Sagang, L.B., da Conceição Bispo, P., Payne, M., Junior, C.S., Favrichon, S., Silgueiro, V., Anderson, L. O., Aragão, L.E.O.C., Fensholt, R., Brandt, M., Ciais, P., Saatchi, S., 2023. Mapping tropical forest degradation with deep learning and Planet NICFI data. Remote Sens. Environ. 298, 113798.
De Bem, P.P., de Carvalho Junior, O.A., Guimaraes, R.F., Trancoso Gomes, R.A., 2020. Change Detection of Deforestation in the Brazilian Amazon Using Landsat Data and Convolutional Neural Networks. Remote Sens. 12 (6), 901.
Decuyper, M., Chávez, R.O., Lohbeck, M., Lastra, J.A., Tsendbazar, N., Hackländer, J., Herold, M., Vågen, T.-G., 2022. Continuous monitoring of forest change dynamics with satellite time series. Remote Sens. Environ. 269, 112829.
Ehrig, C., Cleophas, C., Forestier, G., 2024. The impact of data set similarity and diversity on transfer learning success in time series forecasting. arXiv preprint arXiv: 2404.06198.
Ghamisi, P., Rasti, B., Yokoya, N., Wang, Q., Hofle, B., Bruzzone, L., Bovolo, F., Chi, M. M., Anders, K., Gloaguen, R., Atkinson, P.M., Benediktsson, J.A., 2019. Multisource and multitemporal data fusion in remote sensing: A comprehensive review of the state of the art. IEEE Geosci. Remote Sens. Mag. 7 (1), 6–39.
Grecchi, R.C., Beuchle, R., Shimabukuro, Y.E., Aragão, L.E., Arai, E., Simonetti, D., Achard, F., 2017. An integrated remote sensing and GIS approach for monitoring areas affected by selective logging: A case study in northern Mato Grosso, Brazilian Amazon. Int. J. Appl. Earth Obs. Geoinf. 61, 70–80.
Griffiths, P., Jakimow, B., Hostert, P., 2018. Reconstructing long term annual deforestation dynamics in Para and Mato Grosso using the Landsat archive. Remote Sens. Environ. 216, 497–513.
Guo, Z., Guo, H., Liu, X., Zhou, W., Wang, Y., Fan, Y., 2022. Sar2color: learning imaging characteristics of sar images for sar-to-optical transformation. Remote Sens. 14 (15), 3740.
Hamrouni, Y., Paillassa, E., Chéret, V., Monteil, C., Sheeren, D., 2021. From local to global: A transfer learning based approach for mapping poplar plantations at national scale using Sentinel-2. ISPRS J. Photogramm. Remote Sens. 171, 76–100.
Hansen, M.C., Potapov, P.V., Moore, R., Hancher, M., Turubanova, S.A., Tyukavina, A., Thau, D., Stehman, S.V., Goetz, S.J., Loveland, T.R., 2013. High-resolution global maps of 21st-century forest cover change. Science 342, 850–853.
Hansen, M.C., Krylov, A., Tyukavina, A., Potapov, P.V., Turubanova, S., Zutta, B., Ifo, S., Margono, B., Stolle, F., Moore, R., 2016. Humid tropical forest disturbance alerts using Landsat data. Environ. Res. Lett. 11 (3), 034008.
Hethcoat, M.G., Edwards, D.P., Carreiras, J.M., Bryant, R.G., Franca, F.M., Quegan, S., 2019. A machine learning approach to map tropical selective logging. Remote Sens. Environ. 221, 569–582.
Hethcoat, M.G., Carreiras, J.M.B., Edwards, D.P., Bryant, R.G., Quegan, S., 2021. Detecting tropical selective logging with C-band SAR data may require a time series approach. Remote Sens. Environ. 259, 112411.
Hussain, M., Chen, D., Cheng, A., Wei, H., Stanley, D., 2013. Change detection from remotely sensed images: From pixel-based to object-based approaches. ISPRS J. Photogramm. Remote Sens. 80, 91–106.
Jamali, A., Roy, S.K., Li, J., Ghamisi, P., 2023. TransU-Net + +: Rethinking attention gated TransU-Net for deforestation mapping. Int. J. Appl. Earth Obs. Geoinf. 120, 103332.
Kattenborn, T., Leitloff, J., Schiefer, F., Hinz, S., 2021. Review on Convolutional Neural Networks (CNN) in vegetation remote sensing. ISPRS J. Photogramm. Remote Sens. 173, 24–49.
Kirillov, A., Mintun, Eric, Ravi, Nikhila, Mao, Hanzi, Rolland, Chloe, Gustafson, Laura, Xiao, Tete, Whitehead, Spencer, Berg, Alexander C., Lo, Wan-Yen, Dollár, Piotr, Girshick, R., 2023. Segment anything. In: Proc. IEEE/CVF International Conference on Computer Vision, pp. 4015–4026.
Li, Y., Mao, H., Girshick, R., He, K., 2022a. Exploring plain vision transformer backbones for object detection. In: European conference on computer vision, pp. 280–296.
Li, Y., Xu, X., Wu, Z., Fan, H., Tong, X., Liu, J., 2022b. A forest type-specific threshold method for improving forest disturbance and agent attribution mapping. GISci. Remote Sens. 59 (1), 1624–1642.
Li, J., Han, X., Qin, Y., Tan, F., Chen, Y., Wang, Z., Song, H., Zhou, X., Zhang, Y., Hu, L., Hu, P., 2023. Artificial intelligence accelerates multi-modal biomedical process: A Survey. Neurocomputing 558, 126720.
Lin, X., Xiang, Yangyang, Zhang, Li, Yang, Xin, Zengqiang Yan, L.Y., 2023. SAMUS: Adapting segment anything model for clinically-friendly and generalizable ultrasound image segmentation. arXiv preprint arXiv: 2309.06824.
Liu, X., Wu, Y., Liang, W., Cao, Y., Li, M., 2022. High resolution SAR image classification using global-local network structure based on vision transformer and CNN. IEEE Geosci. Remote Sens, Lett, p. 19.
Liu, C., Sun, Y., Xu, Y., Sun, Z., Zhang, X., Lei, L., Kuang, G., 2024. A review of optical and SAR image deep feature fusion in semantic segmentation. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 17.
Ma, J., He, Y., Li, F., Han, L., You, C., Wang, B., 2024a. Segment anything in medical images. Nat. Commun. 15 (1), 654.
Ma, Y.C., Chen, S., Ermon, S., Lobell, D.B., 2024b. Transfer learning in environmental remote sensing. Remote Sens. Environ. 301, 113924.
Martins, V.S., Roy, D.P., Huang, H., Boschetti, L., Zhang, H., Yan, L., 2022. Deep learning high resolution burned area mapping by transfer learning from Landsat-8 to PlanetScope. Remote Sens. Environ. 280, 113203.
McGregor, I.R., Connette, G., Gray, J.M., 2024. A multi-source change detection algorithm supporting user customization and near real-time deforestation detections. Remote Sens. Environ. 308, 114195.
Miura, S., Amacher, M., Hofer, T., San-Miguel-Ayanz, J., Thackway, R., 2015. Protective functions and ecosystem services of global forests in the past quarter-century. For. Ecol. Manag. 352, 35–46.
Mullissa, A., Reiche, J., Herold, M., 2023. Deep learning and automatic reference label harvesting for Sentinel-1 SAR-based rapid tropical dry forest disturbance mapping. Remote Sens. Environ. 298, 113799.
Ngo, B.H., Do-Tran, N.T., Nguyen, T.N., Jeon, H.G., Choi, T.J., 2024. Learning CNN on ViT: A Hybrid Model to Explicitly Class-specific Boundaries for Domain Adaptation. In: Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 28545–28554.
Pan, S.J., Yang, Q., 2009. A survey on transfer learning. IEEE Trans. Knowl. Data Eng. 22, 1345–1359.
Peng, L., Wei, T., Chen, X., Chen, X., Sun, R., Wan, L., Chen, J., Zhu, X., 2024. Human-annotated label noise and their impact on ConvNets for remote sensing image scene classification. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 18.
Pickens, A.H., Hansen, M.C., Adusei, B., Potapov, P., 2020. Sentinel-2 Forest Loss Alert. In: Global Land Analysis and Discovery (GLAD). University of Maryland.
Qiu, S., Zhu, Z., Olofsson, P., Woodcock, C.E., Jin, S., 2023. Evaluation of Landsat image compositing algorithms. Remote Sens. Environ. 285, 113375.
Reiche, J., Hamunyela, E., Verbesselt, J., Hoekman, D., Herold, M., 2018. Improving near-real time deforestation monitoring in tropical dry forests by combining dense Sentinel-1 time series with Landsat and ALOS-2 PALSAR-2. Remote Sens. Environ. 204, 147–161.
Reiche, J., Mullissa, A., Slagter, B., Gou, Y., Tsendbazar, N.E., Braun, C., Vollrath, A., Weisse, M.J., Stolle, F., Pickens, A., Donchyts, G., Clinton, N., Gorelick, N., Herold, M., 2021. Forest disturbance alerts for the Congo Basin using Sentinel-1. Environ. Res. Lett. 16 (2), 024005.
Reichstein, M., Camps-Valls, G., Stevens, B., Jung, M., Denzler, J., Carvalhais, N., Prabhat, F., 2019. Deep learning and process understanding for data-driven Earth system science. Nature 566, 195–204.
Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., Chen, L.C., 2018. Mobilenetv2: Inverted residuals and linear bottlenecks. In: Proc. IEEE conference on computer vision and pattern recognition, pp. 4510–4520.
Senf, C., Seidl, R., 2021. Mapping the forest disturbance regimes of Europe. Nat. Sustain. 4 (1), 63–70.
Shannon, C.E., 1948. A mathematical theory of communication. Bell Syst. Tech. J. 27, 379–423.
Singh, P., Shree, R., 2016. Analysis and effects of speckle noise in SAR images. In: 2016 2nd International Conference on Advances in Computing, Communication, & Automation (ICACCA)(Fall). IEEE, pp. 1–5.
Slagter, B., Reiche, J., Marcos, D., Mullissa, A., Lossou, E., Peña-Claros, M., Herold, M., 2023. Monitoring direct drivers of small-scale tropical forest disturbance in near real-time with Sentinel-1 and-2 data. Remote Sens. Environ. 295, 113655.
Solorzano, J.V., Francois Mas, J., Alberto Gallardo-Cruz, J., Gao, Y., Fernandez-Montes de Oca, A., 2023. Deforestation detection using a spatio-temporal deep learning approach with synthetic aperture radar and multispectral images. ISPRS J. Photogramm. Remote Sens. 199, 87–101.
Song, Hui Yang, Yanlan, Wu, Zhang, Peng, Wang, Biao, Han, G., 2024. A multispectral remote sensing crop segmentation method based on Segment Anything Model using Multi-stage Adaptation Fine-tuning. IEEE Trans. Geosci. Remote Sens. 62.
Tang, Q., Liang, J., Zhu, F., 2023a. A comparative review on multi-modal sensors fusion based on deep learning. Signal Process. 109165.

Tang, X.J., Bratley, K.H., Cho, K.J., Bullock, E.L., Olofsson, P., Woodcock, C.E., 2023b. Near real-time monitoring of tropical forest disturbance by fusion of Landsat, Sentinel-2, and Sentinel-1 data. Remote Sens. Environ. 294, 113626.
Tsokas, A., Rysz, M., Pardalos, P.M., Dipple, K., 2022. SAR data applications in earth observation: An overview. Expert Syst. Appl. 205, 117342.
Tu, Z., Gu, L., Wang, X., Jiang, B., 2024. Ultrasound SAM Adapter: Adapting SAM for Breast Lesion Segmentation in Ultrasound Images. arXiv preprint arXiv: 2404.14837.
Tuia, D., Volpi, M., Copa, L., Kanevski, M., Munoz-Mari, J., 2011. A survey of active learning algorithms for supervised remote sensing image classification. IEEE J. Sel. Top. Signal Process. 5, 606–617.
Van der Maaten, L., Hinton, G., 2008. Visualizing data using t-SNE. J. Mach. Learn. Res. 9, 2579–2605.
Vega, Juan, P., Ostwald Pedro da Costa, G.A., Feitosa, R.Q., Ortega Adarme, M.X., de Almeida, C.A., Heipke, C., Rottensteiner, F., 2021. An unsupervised domain adaptation approach for change detection and its application to deforestation mapping in tropical biomes. ISPRS J. Photogramm. Remote Sens. 181, 113–128.
Wan, L., Liu, Y.F., He, Y., Cen, H.Y., 2023. Prior knowledge and active learning enable hybrid method for estimating leaf chlorophyll content from multi-scale canopy reflectance. Comput. Electron. Agric. 214, 108308.
Wang, Y.X., Ziv, G., Adami, M., Mitchard, E., Batterman, S.A., Buermann, W., Marimon, B.S., Marimon Jr., B., Reis, S.M., Rodrigues, D., Galbraith, D., 2019. Mapping tropical disturbed forests using multi-decadal 30 m optical satellite imagery. Remote Sens. Environ. 221, 474–488.
Wang, J., Yang, D., Chen, S., Zhu, X., Wu, S., Bogonovich, M., Guo, Z., Zhu, Z., Wu, J., 2021a. Automatic cloud and cloud shadow detection in tropical areas for PlanetScope satellite images. Remote Sens. Environ. 264, 112604.
Wang, S., Li, C., Wang, R., Liu, Z., Wang, M., Tan, H., Wu, Y., Liu, X., Sun, H., Yang, R., 2021b. Annotation-efficient deep learning for automatic medical image segmentation. Nat. Commun. 12, 5915.
Weiss, K., Khoshgoftaar, T.M., Wang, D., 2016. A survey of transfer learning. J. Big Data 3, 1–40.
Wu, D.R., Lin, C.T., Huang, J., 2019. Active learning for regression using greedy sampling. Inf. Sci. 474, 90–105.
Yan, Z.Y., Li, J.X., Li, X.X., Zhou, R.X., Zhang, W.K., Feng, Y.C., Diao, W.H., Fu, K., Sun, X., 2023. RingMo-SAM: A Foundation Model for Segment Anything in Multimodal Remote-Sensing Images. IEEE Trans. Geosci. Remote Sens. 61.
Zhang, J., Yang, X., Jiang, R., Shao, W., Zhang, L., 2024. RSAM-Seg: A SAM-based Approach with Prior Knowledge Integration for Remote Sensing Image Semantic Segmentation. arXiv preprint arXiv: 2402.19004.
Zhao, F., Sun, R., Zhong, L., Meng, R., Huang, C., Zeng, X., Wang, M., Li, Y., Wang, Z., 2022. Monthly mapping of forest harvesting using dense time series Sentinel-1 SAR imagery and deep learning. Remote Sens. Environ. 269, 112822.
Zhou, J., Chen, J., Chen, X., Zhu, X., Qiu, Y., Song, H., Rao, Y., Zhang, C., Cao, X., Cui, X., 2021. Sensitivity of six typical spatiotemporal fusion methods to different influential factors: A comparative study for a normalized difference vegetation index time series reconstruction. Remote Sens. Environ. 252, 112130.
Zhou, X.C., Liang, F., Chen, L.H., Liu, H.J., Song, Q.Q., Vivone, G., Chanussot, J., 2024. MeSAM: Multiscale Enhanced Segment Anything Model for Optical Remote Sensing Images. IEEE Trans. Geosci, Remote Sens, p. 62.
Zhu, L.J., Dai, J.J., Liu, Y., Yuan, S.S., Qin, T.L., Walker, J.P., 2024. A cross-resolution transfer learning approach for soil moisture retrieval from Sentinel-1 using limited training samples. Remote Sens. Environ. 301, 113944.
Zhuang, F., Qi, Z., Duan, K., Xi, D., Zhu, Y., Zhu, H., Xiong, H., He, Q., 2020. A comprehensive survey on transfer learning. In Proc. IEEE 109, 43–76.